NLP PRACTICAL EXERCISE - 05

Aim:

To implement **Word Sense Disambiguation (WSD)** using the **Simplified Lesk algorithm** to identify the correct meaning of an ambiguous word based on its context.

Procedure :

Step 1 — Open Google Colab

Step 2 — Import NLTK

CODE - import nltk

Step 3 — Download WordNet

CODE -

```
nltk.download('wordnet')
nltk.download('omw-1.4')
nltk.download('punkt')
nltk.download('punkt_tab')
```

Step 4 — Import required libraries

CODE -

```
from nltk.corpus import wordnet as wn
from nltk.tokenize import word_tokenize
```

Step 5 — Define the Simplified Lesk function

CODE -

```
def simplified_lesk(sentence, word):
    context = set(word_tokenize(sentence.lower()))
    best_sense = None
    max_overlap = 0
    for sense in wn.synsets(word):
        definition = set(word_tokenize(sense.definition().lower()))
        overlap = len(context.intersection(definition))
        if overlap > max_overlap:
            max_overlap = overlap
            best_sense = sense
```

```
    return best_sense, max_overlap
```

Step 6 — Get user input
CODE -

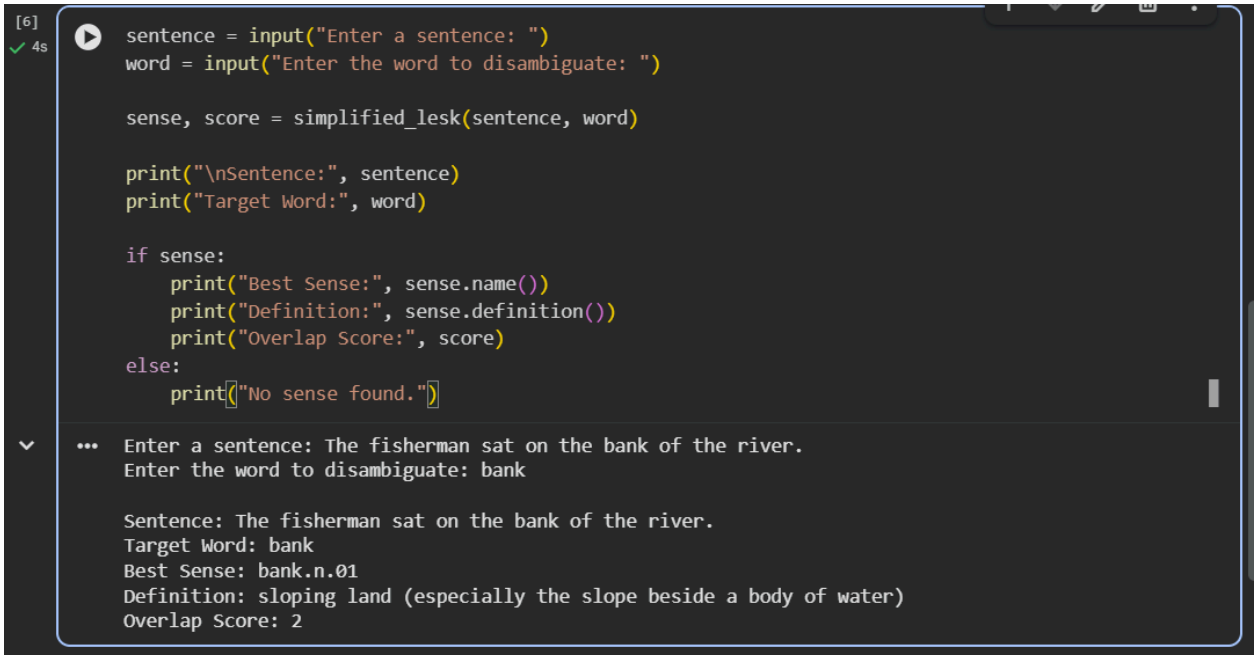
```
sentence = input("Enter a sentence: ")
word = input("Enter the word to disambiguate: ")
sense, score = simplified_lesk(sentence, word)
print("\nSentence:", sentence)
print("Target Word:", word)
if sense:
    print("Best Sense:", sense.name())
    print("Definition:", sense.definition())
    print("Overlap Score:", score)
else:
    print("No sense found.")
```

Step 7 — Test the practical
Enter a sentence: I went to the bank to deposit money.
Enter the word to disambiguate: bank

Result:

Thus, **Word Sense Disambiguation was successfully implemented** using the Simplified Lesk algorithm, and the appropriate sense of the given word was identified based on the context.

EXPECTED OUTPUT : (SAMPLE)



```
[6]
✓ 4s

sentence = input("Enter a sentence: ")
word = input("Enter the word to disambiguate: ")

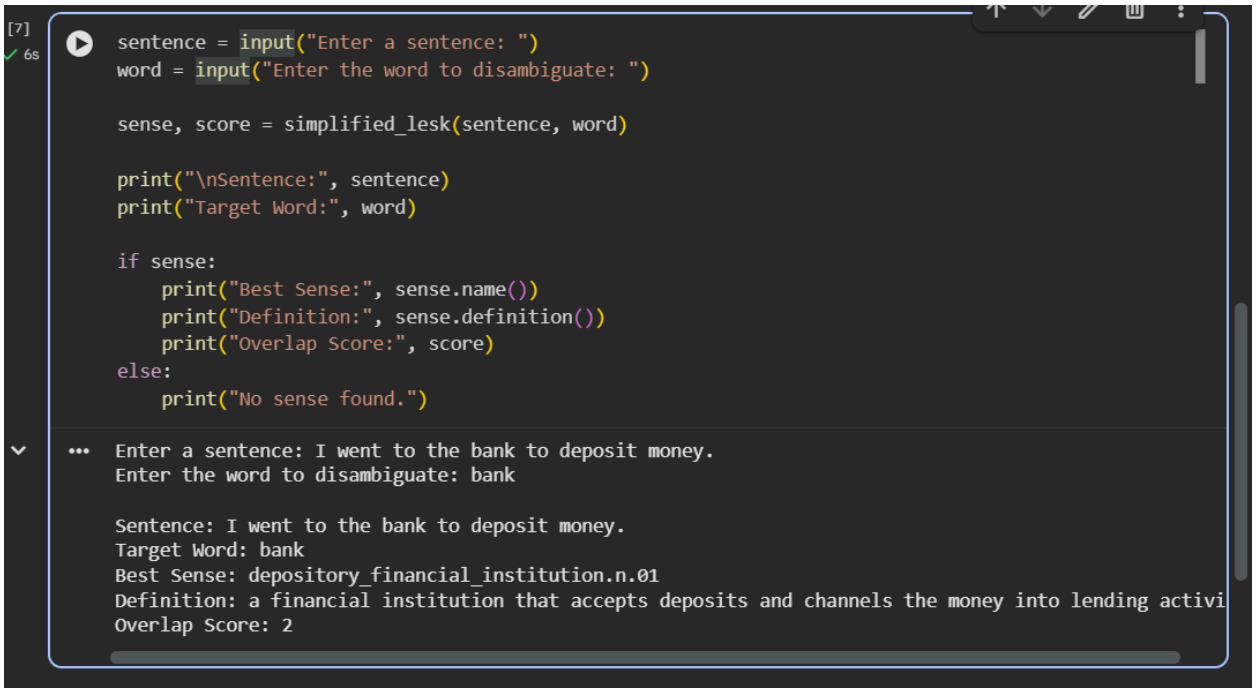
sense, score = simplified_lesk(sentence, word)

print("\nSentence:", sentence)
print("Target Word:", word)

if sense:
    print("Best Sense:", sense.name())
    print("Definition:", sense.definition())
    print("Overlap Score:", score)
else:
    print("No sense found.")

Enter a sentence: The fisherman sat on the bank of the river.
Enter the word to disambiguate: bank

Sentence: The fisherman sat on the bank of the river.
Target Word: bank
Best Sense: bank.n.01
Definition: sloping land (especially the slope beside a body of water)
Overlap Score: 2
```



```
[7]
✓ 6s

sentence = input("Enter a sentence: ")
word = input("Enter the word to disambiguate: ")

sense, score = simplified_lesk(sentence, word)

print("\nSentence:", sentence)
print("Target Word:", word)

if sense:
    print("Best Sense:", sense.name())
    print("Definition:", sense.definition())
    print("Overlap Score:", score)
else:
    print("No sense found.")

Enter a sentence: I went to the bank to deposit money.
Enter the word to disambiguate: bank

Sentence: I went to the bank to deposit money.
Target Word: bank
Best Sense: depository_financial_institution.n.01
Definition: a financial institution that accepts deposits and channels the money into lending activi
Overlap Score: 2
```